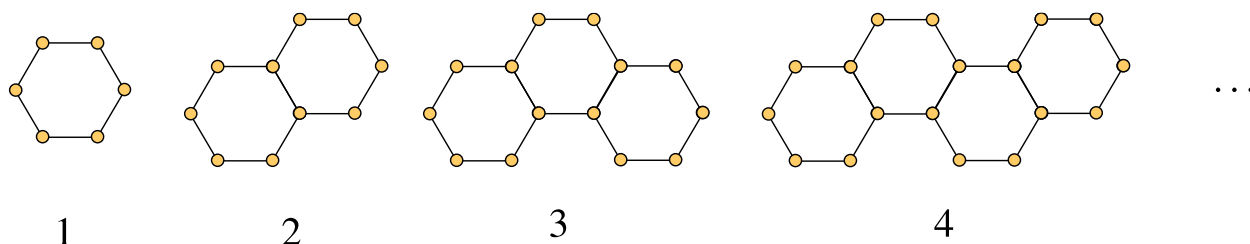


Worksheet 1: Rules hidden in molecular patterns (Stage 4 Mathematics)

The following are the first four of a **sequence of growing patterns** consisting of dots



Hexagons (n)	1	2	3	4	5	15	?	n
Dots (d)	6	10	14	?	?	?	402	?

Questions:

- Describe with words how to draw/construct a member of the family from the previous pattern

For each step, we place a new hexagon next to the rightmost hexagon of the previous pattern. The two hexagons **share a common edge**.

Side note: we can actually be more precise than this, but for that we will need to label edges of hexagons and introduce the notion of translation as a transformation in geometry

- How many dots are there for 4 hexagons? What about 5 hexagons?

For the pattern with 4 hexagons: **18 dots**. For the pattern with 5 hexagons: **22 dots**

- How many hexagons should there be in order to have 402 dots?

With each new pattern in the family, we add 4 more dots. This can be seen from the way number of dots change from when there are 4 hexagons to when there are 5 hexagons. We can also confirm this by looking at the description in the answer to the first question, each new addition of a hexagon would bring us 6 new dots without the common edge and $6-2=4$ dots with the common edge.

Now $402-22=380$, which is 4×95 , so there are 95 more hexagons than when there are 5, that is **100 hexagons**.

- Write down a formula that gives number of dots for any given number of hexagons? Can you justify your answer?

4n+2. There are a number of ways we can arrive at this answer. In the beginning, there is 1 hexagon with 6 dots. At each step, there are 4 more dots. Therefore, when there are n hexagons, there has been n-1 additions and we get $6 + 4(n-1) = 4n+2$.

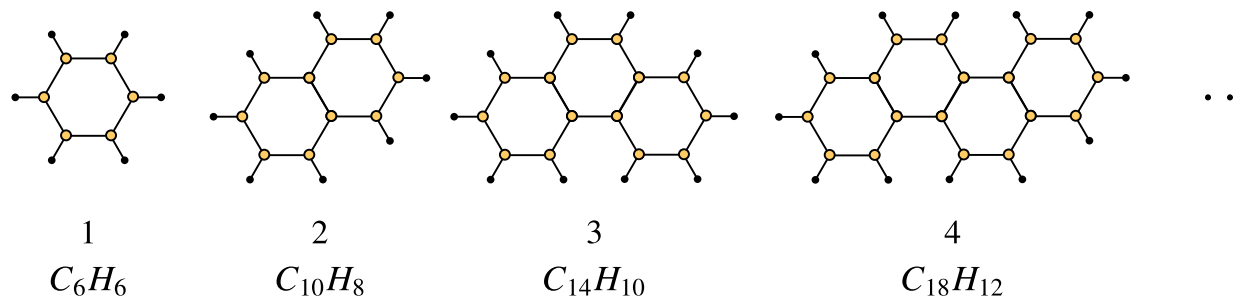
Alternatively, when there are n hexagons with no shared edges, there are 6n dots. But for a pattern in the family with n hexagons, there are n-1 shared edges, which gives $6n - 2(n-1) = 4n+2$ dots again.

- (Advanced) Are there a pattern in this family with precisely 40 dots? Can you describe a procedure that recover the number of hexagons given the number of dots?

We can solve the equation $4n+2=40$ to get $n=9.5$. This is not a whole number. No whole number n will solve the equation, so there is no pattern in this family with precisely 40 dots. Alternatively: suppose we have a pattern in the family with exactly 40 dots, we should be able to strip away 4 dots each time to reduce it to a single hexagon with just 6 dots. We get: 40, 36, 32, ..., 8, 4, before reaching 0. We never get 6 dots. This means that no pattern in our family has exactly 40 dots.

Worksheet 2: Rules hidden in molecular patterns (Stage 4 Mathematics)

The following are structures of a family of hydrocarbon molecules, yellow dots represent carbon (C) atoms, black dots represent hydrogen (H) atoms. Molecular formula C_6H_6 means there are 6 carbon and 6 hydrogen atoms.



Hexagons (n)	1	2	3	4	15	?	?
C atoms (c)	6	10	14	18	?	402	?
H atoms (h)	6	8	10	12	?	?	26
Molecular formula	C_6H_6	$C_{10}H_8$	$C_{14}H_{10}$	$C_{18}H_{12}$	$C_?H_?$	$C_{402}H_?$	$C_?H_{26}$

Questions:

- A molecule C_nH_m in this family has 15 hexagons, find the number of C and H atoms.

For a molecule in the family with n hexagons, there are $4n+2$ C-atoms and **$2n+4$ H-atoms**.

The count of H atoms is given by this process: we observe that at each step when a new hexagon is added to the previous pattern, the two H-atoms attached to the two C-atoms that form the common edge is gone, and 4 new C-atoms are each attached to a new H-atoms. So there are in total 2 new H-atoms added at each step. In the beginning, there are 6 H-atoms. A molecule with n hexagons in this family is arrived at after $(n-1)$ attachments of hexagons, therefore the number of H-atoms is $6+2(n-1)=2n+4$.

Note:

1. Above answers the last question
2. There are other ways to derive the same formula.

Using this formula, for 15 hexagons, there are $4n+2=4\times 15+2=$ **62 C-atoms** and $2n+4=2\times 15+4=$ **34 H-atoms**. The molecular formula is $C_{62}H_{34}$.

- The molecule $C_{402}H_m$ is a member of this family, how many H atoms does it have?

Setting $4n+2=402$, we get $n=100$, there are 100 hexagons in the molecule. Correspondingly, $2n+4=204$. There should be **204 H-atoms** in this molecule.

- The molecule C_nH_{26} is a member of this family, how many hexagons does the molecule has?

Setting $2n+4=26$, we have $2n=24$ by subtracting 4 from both sides. Therefore, $n=12$, there are **12 hexagons** in the molecule.

- A hydrocarbon molecule has the form $C_{22}H_{12}$. Could it be in the family? Why or why not? What about $C_{48}H_{24}$?

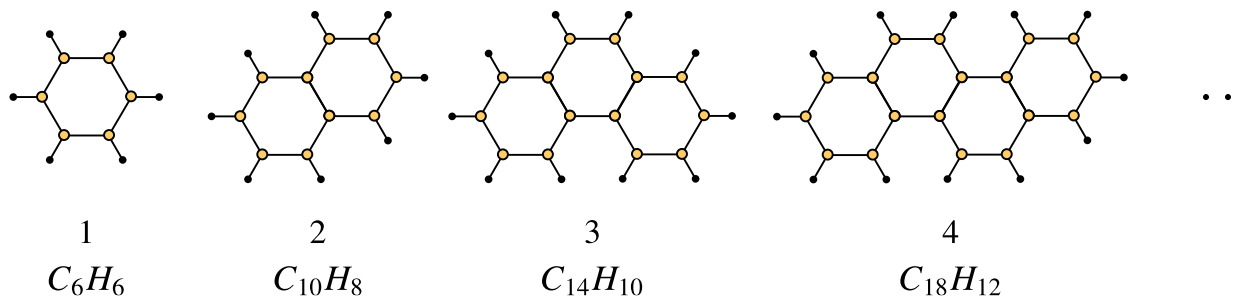
$C_{22}H_{12}$ cannot be in the family because for $4n+2=22$ we get $4n=20$ and $n=5$, but $2n+4=14$ not 12. $C_{48}H_{24}$ cannot be in the family because for $4n+2=48$, we get $4n=46$ which has no whole-number solution.

- For each value of n , can you find a formula that give number of H atoms of a molecule in the family with n hexagons? Justify your answer.

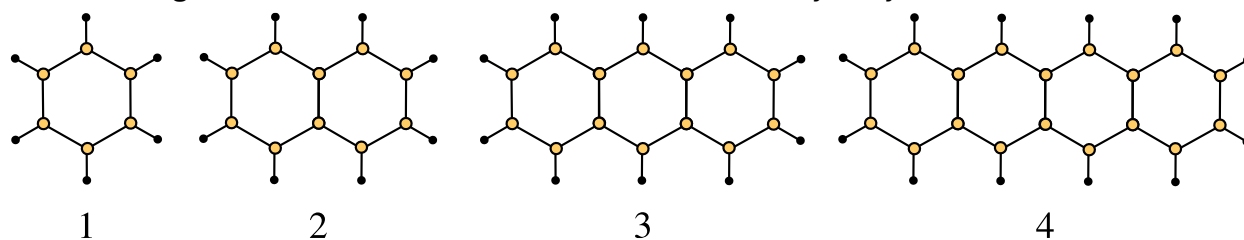
See question 1.

Worksheet 3: Rules hidden in molecular patterns (Stage 4 Mathematics)

In previous worksheets, we considered this family of hydrocarbon molecules:



The following shows the first few members of another family of hydrocarbon molecules:



Task:

- Without making a table of counts of hexagons, C and H atoms, using justification you found of the formulas for counts of C and H atoms in previous family, find a formula for corresponding counts of C and H in this family
- Are the molecular formulas different or the same? Why?

For a molecule with n hexagons in this family, there are $4n+2$ C-atoms and $2n+4$ H-atoms, exactly the same with the previous family.

To see this, we notice that

1. The first two member of both families are actually identical molecules, with identical numbers of C-atoms and H-atoms. These are C_6H_6 and $C_{10}H_8$.
2. The attachment of each new hexagon in either families involves the same increase in the number C-atoms (4 each time) and H-atoms (2 each time)
3. Therefore, there is no difference in the derivation of the formulas. Hence the formulas should be identical.